

Vol 1 Chapter 11 — Observables Reference

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2026-05-23 Saturday

Chapter 11 — Observables Reference

The volume has built the substrate framework for quantum field theory: Hilbert space, integers, discrete symmetries, Casimir algebra, operator zoo, dynamics, gauge theory, scattering, renormalization. This chapter consolidates what the apparatus produces — over six hundred substrate-derived observables, tier-classified, with cross-references to the computational verifications.

The chapter does not derive new content. It indexes.

11.1 Headline derivations

Observable	BST formula	Tier	Match	Where
m_p/m_e	$6\pi^5 = 1836.118$	D	0.002%	Vol 2 Ch 6 (T187)
$1/\alpha$	$N_{\max} =$ $N_C^3 \cdot n_C + \text{rank} = 137$	D	0.026%	Paper #104
Newton's G	$\hbar c (6\pi^5)^2 \alpha^{24} / m_e^2$	D	0.07%	Vol 4 Ch 1 (T1296)
Λ/M_{Pl}^4	$7e^{-282}$	D	0.076 dex	Vol 4 Ch 4 (T1485)
$\sin^2 \theta_W$	$N_C/c_3 = 3/13$	D	0.19%	Vol 2 Ch 2
a_e (electron $g - 2$)	$\alpha/(2\pi) + \text{substrate}$	D	ppt	Vol 2 Ch 8 (K92)
CMB n_s	$1 - n_C/N_{\max} =$ 0.9635	D	0.3 σ	Vol 4 Ch 6
Dark-matter/baryon	$(3n_C + 1)/N_C = 16/3$	D	0.58%	Vol 4 Ch 10

Each is a substrate-mechanical derivation from BST primaries with no fitted parameters. Match precisions span 0.002% to 0.58%, all D-tier.

11.2 Reproduction suite

The complete computational verification is `play/verify_bst.py` — evaluating fifty substrate-derived predictions against measurement. Current status: 49 of 50 PASS at $\leq 1\%$, single warning is an honest-scope flag.

Toy 541 — “five integers to everything” — derives 51 quantities from the BST primaries, 16/16 sub-checks at machine precision. Three-second entry-point verification.

Catalog: over 600 substrate-significant quantities in `data/bst_constants.json` and `data/bst_geometric_invariants.json`, each with evaluable `formula_code`.

11.3 Tier counts

- D-tier: ~130 entries (mechanism closed, match below 1%)

- I-tier: ~230 (match below 1%, mechanism plausible-pending)
- C-tier: ~80 (conditional on external conjectures)
- S-tier: ~160 (qualitative or above 2%)

Total ~600. Tier discipline per Volume 0 Chapter 10: no overclaiming, near misses get scrutiny, honest scope preserved.

11.4 Falsifiers

Five-Absence Prediction Set (Volume 1 Chapter 8 §8.5):

1. No grand unified theory observable phenomena.
2. No proton decay above structural lifetime.
3. No magnetic monopoles.
4. No sterile neutrinos.
5. No supersymmetric partner particles.

Plus: CPT violation at any precision refutes substrate $SO_0(5, 2)$; free quark observation refutes color-confinement topology; CMB n_s measurement diverging from $1 - n_C/N_{\max}$ refutes substrate cosmological boundary condition.

Framework is set up to be falsifiable across multiple independent axes.

11.5 What comes next

Volume 2 begins particle-physics applications in detail. Chapter 1 introduces; Chapters 2–12 take the major Standard Model sectors through their substrate-mechanism derivations. The proton-to-electron mass ratio (Vol 2 Ch 6, already rewritten as recruiter) is one of many.

Where to look this up: `data/bst_constants.json` and `data/bst_geometric_invariants.json` for the complete catalog. `play/verify_bst.py` for the reproduction script. `play/toy_bst_explorer.py` for the interactive REPL. Toy 541 for the entry-point three-second verification. `notes/Five_Absence_Predictions_Principle.md` for the falsifier set. `notes/BST_AC_Theorem_Registry.md` for per-theorem provenance.